

Take-Home Evaluation Components: Submission and Evaluation Guidelines for CS G653/ SS G653

Software Design and Development (team-based; team-size – 1 to 4)

Submission Guidelines

1. Abstract Submission

- The abstract must clearly define the problem statement, objectives, system scope, and key functionalities.
- It should justify the relevance of the proposed system and its suitability for architectural comparison.
- The abstract is subject to instructor approval before implementation begins.

2. Architectural Implementations

- The software system must be fully implemented using **one architectural pattern**.
- The pattern can be chosen freely by a team. Certain constraints would be specified after abstract submission (check the excel sheet)

3. Code and Documentation

- Complete, well-structured, and executable source code must be submitted for the implementation.
- Comprehensive architectural documentation must be prepared for the pattern using the techniques taught in the course (e.g., views, diagrams, rationale, and quality attribute considerations). It should contain a link to the git repository.
- Clear instructions for setup, execution, and testing must be included.

4. Final Submission

- All deliverables (code, documentation, and abstract) must be submitted within the prescribed deadline in the specified format.

Evaluation Scheme

Architectural Pattern Implementation – 30 Marks

Criterion	Marks
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Correct application of the chosen architectural pattern	8
Fulfilment of functional requirements	6
Architectural design quality and adherence to principles	8
Quality and completeness of architectural documentation	6
Code quality, modularity, and maintainability	2
Total	30

Overall Assessment Focus

- Depth of architectural understanding
- Justification of design decisions and trade-offs (architectural tactics)
- Consistency between design, documentation, and implementation
- Professional quality of code and documentation

Research Summary (individual)

Submission Guidelines

1. Paper Selection

- Students must select **one research paper** from the provided pool on a **first-come, first-served basis**.
- Once selected, the paper cannot be chosen by another candidate.

2. Literature Collection

- Students must identify **additional related research papers** using the **snowballing technique** (backward and forward reference tracking).
- All collected papers must be relevant to the chosen research theme.

3. Report Preparation

- The research summary must be written in the **approved LaTeX template** only.
- Length: **4–6 pages**, excluding references.
- The report should include: introduction, literature synthesis, comparative analysis, key insights, and conclusion.

4. Originality and Ethics

- A **plagiarism report** must be submitted along with the final document.
- Similarity index must be **below 20%** (excluding references).

5. Submission Artifacts

- PDF of the final report
- Plagiarism report

Evaluation Rubrics / Scheme (Total: 10 Points)

Criterion	Description	Marks
Understanding of the Selected Paper	Clear articulation of the research problem, objectives, methodology, and key contributions	2
Literature Coverage and Relevance	Appropriateness and relevance of additional papers identified through snowballing	2
Critical Analysis and Synthesis	Depth of comparison, integration of ideas, and synthesis across multiple studies	3
Report Organization and Technical Writing	Logical structure, clarity of expression, academic tone, and adherence to LaTeX template	2
Originality and Academic Integrity	Proper citations and plagiarism similarity below 20%	1

Notes:

- Superficial summaries without synthesis will receive limited marks.
- Excessive plagiarism or non-adherence to format will result in penalties.

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